Eventual minimizers of the adjacency spectral gap in trees and connected graphs

Abstract

The adjacency spectral gap of a connected graph is the difference between its two largest adjacency eigenvalues. We prove that, for every sufficiently large order, every tree minimizing this gap is a symmetric double comet, and every connected graph minimizing it is a symmetric double kite. These results establish the eventual forms of the conjectures of Jovović, Koledin and Stanić for trees and of Stanić for connected graphs. The number of leaves at each terminal vertex of a minimizing tree, and the order of each terminal clique of a minimizing graph, are asymptotic to $n/(2\log n)$. The proofs combine resistance bounds with comparisons against explicit families. For trees, nodal components identify two dominant vertices, and a Schur complement forces their degrees to agree before an exact path comparison eliminates all other branches. For graphs, a centered second eigenvector gives a constant vertex budget, from which two dense cores are extracted. A comparison at fixed order removes exceptional vertices and multiple attachments; a uniform completion argument forces both cores to be cliques. All auxiliary estimates needed for these reductions are proved.

Keywords. Adjacency spectral gap; extremal graphs; trees; double comets; double kites; Perron vector; Schur complement.
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1 Introduction

1.1 Notation and the extremal problems

All graphs are finite, simple and undirected. The order of a graph G is $n = |V(G)|$, and its adjacency matrix is $A(G)$. We write its eigenvalues as

$$\lambda_1(G) \geq \lambda_2(G) \geq \cdots \geq \lambda_n(G).$$

For a connected graph of order at least two, its *adjacency spectral gap* is

$$\delta(G) = \lambda_1(G) - \lambda_2(G) > 0.$$

Whenever a graph is fixed, we suppress G in its eigenvalues. Throughout, $\mathbf{x}$ and $\mathbf{y}$ denote orthonormal real eigenvectors for λ_1 and λ_2, respectively, with $\mathbf{x} > 0$. Their coordinates at a vertex u are x_u and y_u. Auxiliary vectors receive different letters. For any graph H, $\rho(H) = \lambda_1(H)$ denotes its spectral radius. Matrix norms are Euclidean operator norms, and vector norms without a subscript are Euclidean norms. The notation $M \preceq N$ means that $N - M$ is positive semidefinite. All logarithms are natural.

Let $\mathcal{T}_n$ and $\mathcal{G}_n$ be the sets of trees and connected graphs of order n, and put

$$\tau_n = \min_{T \in \mathcal{T}_n} \delta(T), \qquad \eta_n = \min_{G \in \mathcal{G}_n} \delta(G).$$

The minima exist because the sets of isomorphism classes are finite. Write P_j for a path on j vertices and K_j for a complete graph on j vertices. We use *edge length* for every joining path:

- $\mathsf{C}(k, L)$ is the tree obtained from a path of $L \geq 1$ edges by attaching $k \geq 0$ new leaves to each endpoint. It is a symmetric double comet and has $2k + L + 1$ vertices.
- For $k \geq 1$, $\mathsf{K}(k, L)$ consists of two disjoint copies of K_k, with one selected vertex in each joined by a path of $L \geq 1$ edges whose internal vertices are new. It is a symmetric double kite and has $2k + L - 1$ vertices.

Thus $\mathsf{C}(k, L)$ is $C(k, L+1)$ in the notation of Jovović et al. [2], whereas $\mathsf{K}(k, L)$ is $DK(k, L-1)$ in the notation of Stanić [6].

1.2 Previous work

Stanić formulated the conjecture that some double kite attains η_n at every order [6, Conjecture 2.1]. That paper verified the conjecture by exhaustive computation through order ten, reported counterexample searches through order fifteen, and derived equations for the first two eigenvalues of double kites, together with spectral-gap bounds and structural restrictions on possible minimizers. In particular, its double-kite calculations exhibit the role of the long joining path.

Jovović, Koledin and Stanić proposed the corresponding conjecture that some symmetric double comet attains τ_n [2, Conjecture 2.1]. They verified it through order twenty. They also proved that, for a fixed internal path, moving one terminal leaf from the larger end of a generalized double comet to the smaller end decreases its gap whenever the two leaf counts differ by at least two [2, Theorem 2.3]. Their further results include eigenvalue equations and structural conditions on extremal trees. The two conjectures are included in the survey of Liu and Ning [5].

Several related extremal problems concern λ_2 itself. Kumar, Mohar, Pragada and Zhan determine the trees maximizing the second adjacency eigenvalue at prescribed order and diameter, and develop spectral-center decompositions for the associated extremal problems [3]. The same authors study the maximum of nonnegative convex combinations of the first two eigenvalues over trees of fixed order [4]. Wang and Guo investigate extremal second eigenvalues in classes of graphs at prescribed order [7]. These problems provide related structural settings; the objective here is the difference $\lambda_1 - \lambda_2$, and all comparisons needed for that objective are established below.

1.3 Main results and proof strategy

Our first theorem identifies every minimizing tree at sufficiently large orders.

Theorem 1.1. *There is an integer $N_{\mathcal{T}}$ such that, for every $n \geq N_{\mathcal{T}}$, every $T \in \mathcal{T}_n$ with $\delta(T) = \tau_n$ is isomorphic to $\mathsf{C}(k, L)$ for some $2k + L + 1 = n$. Along every sequence of such minimizers,*

$$k \sim \frac{n}{2 \log n}, \qquad L = n - (1 + o(1)) \frac{n}{\log n}.$$

The second theorem gives the corresponding conclusion for connected graphs.

Theorem 1.2. *There is an integer $N_{\mathcal{G}}$ such that, for every $n \geq N_{\mathcal{G}}$, every $G \in \mathcal{G}_n$ with $\delta(G) = \eta_n$ is isomorphic to $\mathsf{K}(k, L)$ for some $2k + L - 1 = n$. Along every sequence of such minimizers,*

$$k \sim \frac{n}{2 \log n}, \qquad L = n - (1 + o(1)) \frac{n}{\log n}.$$

Neither theorem asserts uniqueness of the minimizing integer k. The conclusions concern all sufficiently large orders; the conjectures at every order are not consequences of these statements. No explicit threshold is needed in the proofs.

Both arguments begin by comparing a resistance lower bound with a symmetric comparison family. For trees, this determines the diameter and spectral scales. Two sign components of $\mathbf{y}$ then produce vertices whose degrees exhaust the branching budget to leading order. Deleting these vertices leaves a matrix of operator norm $o(\lambda_1)$. A two-dimensional Schur complement places the two integer degrees in a common interval of length $o(1)$, forcing equality. An exact path-resolvent estimate then shows that any remaining branch loses to a double comet of the same order.

For general graphs we center $\mathbf{y}$ by subtracting a multiple of $\mathbf{x}$. The centered extrema and an exact path-resistance identity leave at most two vertices of slack in a neighborhood count. All large positive and negative coordinates form two dense terminal cores, with a bounded residual attachment set. A finite-rank resolvent estimate quantifies the spectral benefit of these attachments, while path transmission quantifies their effect on the gap. The comparison at fixed order removes every extra vertex, every unequal core order, and every multiple attachment. Finally, a uniform rooted-core comparison shows that any missing clique edge strictly increases the gap.

We use the Perron–Frobenius theorem, the variational principle, eigenvalue interlacing and elementary symmetric-matrix inequalities in their standard forms; see Brouwer and Haemers [1]. The path identities, resistance estimates, and Schur-complement comparisons specific to this problem are proved in full. Asymptotic statements are always made along an arbitrary sequence of minimizers whose orders tend to infinity. A contradiction to an arbitrary sequence of exceptions therefore establishes one threshold valid for every minimizer.

2 Resistance, path resolvents and two-vertex elimination

2.1 The ground-state identity

Put $\pi_u = x_u^2$, so that $\sum_u \pi_u = 1$. For a real function f on $V(G)$, set $z_u = x_u f_u$. The Perron equations give

$$\mathbf{z}^T(\lambda_1 I - A)\mathbf{z} = \sum_{uv\in E(G)} x_u x_v (f_u - f_v)^2. \tag{2.1}$$

Indeed, expanding the right-hand side and collecting the coefficient of f_u^2 gives $x_u \sum_{v\sim u} x_v = \lambda_1 x_u^2$. The variational principle consequently yields

$$\delta(G) = \min_{\substack{f\neq 0 \\ \sum_u \pi_u f_u = 0}} \frac{\sum_{uv\in E(G)} x_u x_v (f_u - f_v)^2}{\sum_u \pi_u f_u^2}. \tag{2.2}$$

For weighted mean-zero f, direct expansion also gives

$$\sum_u \pi_u f_u^2 = \sum_{u<v} \pi_u \pi_v (f_u - f_v)^2, \tag{2.3}$$

where any fixed ordering of the vertices can be used.

Lemma 2.1. *If G is connected, has order $n \geq 3$ and diameter D, then*

$$\delta(G) \geq \frac{2}{D(n-1)\lambda_1^{D-1}}.$$

Proof. For each unordered pair u, v, choose a shortest path Q_{uv} and put

$$R_{uv} = \sum_{ab\in E(Q_{uv})} \frac{1}{x_a x_b}.$$

Cauchy–Schwarz on this path gives $(f_u - f_v)^2 \le R_{uv}\mathcal{E}(f)$, where $\mathcal{E}(f)$ is the numerator in (2.2). If ab is oriented toward v on an h-edge u–v path, successive Perron equations imply $x_u x_v/(x_a x_b) \le \lambda_1^{h-1}$. Since $\lambda_1 \ge 1$,

$$\sum_{u<v} x_u^2 x_v^2 R_{uv} \le D\lambda_1^{D-1} \sum_{u<v} x_u x_v \le \frac{D(n-1)\lambda_1^{D-1}}{2}.$$

The last inequality follows from $\|\mathbf{x}\|_1^2 \le n$. Now apply (2.3) and (2.2). □

2.2 Exact path identities

Define $\phi_j(z) = \det(zI - A(P_j))$ for $j \ge 0$, with $\phi_0 = 1$ and $\phi_{-1} = 0$. Expansion at a path endpoint gives $\phi_j = z\phi_{j-1} - \phi_{j-2}$. For $z > 2$, write

$$r(z) = \frac{z - \sqrt{z^2-4}}{2} \in (0,1), \qquad z = r(z) + r(z)^{-1}.$$

Lemma 2.2. *For $j \ge 0$ and $z > 2$,*

$$\phi_j(z) = \frac{r(z)^{-j-1} - r(z)^{j+1}}{r(z)^{-1} - r(z)}, \qquad 0 < \phi_j(z) \le z^j.$$

For a path with $L \ge 2$ edges, the endpoint diagonal entries and the opposite-corner entry of its internal resolvent $(zI - A(P_{L-1}))^{-1}$ are

$$s_L(z) = \frac{\phi_{L-2}(z)}{\phi_{L-1}(z)} = r(z)\frac{1 - r(z)^{2L-2}}{1 - r(z)^{2L}}, \tag{2.4}$$

$$h_L(z) = \frac{1}{\phi_{L-1}(z)} = \frac{(1 - r(z)^2)r(z)^{L-1}}{1 - r(z)^{2L}}. \tag{2.5}$$

In particular, $s_L(z) - r(z) = -h_L(z)r(z)^L$.

Proof. The formula for ϕ_j satisfies its recurrence and initial values. Its positivity also follows from $z > \rho(P_j)$; the upper bound follows inductively from the recurrence. Cofactor expansion gives the two stated resolvent entries. Substitution of the formula for ϕ_j proves (2.4)–(2.5) and their final identity. □

The expression in (2.5) will also be denoted by $h_1(z) = 1$ when $L = 1$; in that case it represents a direct edge.

Lemma 2.3 (Centering and exact resistance)**.** *There exists $c \in \mathbb{R}$ such that $\mathbf{w} = \mathbf{y} - c\mathbf{x}$ satisfies $\max_u w_u = M = -\min_u w_u > 0$. It has the properties*

$$\|\mathbf{w}\|^2 = 1 + c^2, \quad M^2 \ge \frac{1}{n}, \quad |c|/M \le \sqrt{n}, \quad (\lambda_2 I - A)\mathbf{w} = c\delta\mathbf{x}. \tag{2.6}$$

If $w_u > 0 > w_v$ and $h = \operatorname{dist}(u,v)$, then

$$\delta(G) \ge \frac{4w_u|w_v|}{\phi_{h-1}(\lambda_1)}. \tag{2.7}$$

Proof. The continuous function $c \mapsto \max_u(y_u - cx_u) + \min_u(y_u - cx_u)$ is strictly decreasing and tends to opposite infinities at the two ends of the real line. Its zero gives the asserted centering. Here $M > 0$, since otherwise $\mathbf{y}$ would be a multiple of $\mathbf{x}$. Orthogonality gives

$\|\mathbf{w}\|^2 = 1 + c^2$ and $\mathbf{x}^T\mathbf{w} = -c$. Thus $nM^2 \geq 1$ and $|c| \leq M\|\mathbf{x}\|_1 \leq M\sqrt{n}$. The last equation in (2.6) is obtained by applying $\lambda_2 I - A$.

Choose a shortest path $u = v_0, \ldots, v_h = v$. Set $a_0 = x_u$, $a_h = x_v$ and solve $\lambda_1 a_i = a_{i-1} + a_{i+1}$ in its interior. The inverse of $\lambda_1 I - A(P_{h-1})$ is nonnegative: the internal path is a proper principal subgraph of the connected graph, so its spectral radius is strictly below λ_1. The Perron equations, with all contributions off the path retained as nonnegative forcing, imply $x_{v_i} \geq a_i > 0$. Put $b_0 = 0$ and $b_i = \phi_{i-1}(\lambda_1)$ for $i \geq 1$. The shared recurrence implies

$$a_i b_{i+1} - a_{i+1} b_i = x_u \qquad (0 \leq i < h).$$

Divide by $a_i a_{i+1}$ and telescope to obtain

$$\sum_{i=0}^{h-1} \frac{1}{x_{v_i} x_{v_{i+1}}} \leq \sum_{i=0}^{h-1} \frac{1}{a_i a_{i+1}} = \frac{\phi_{h-1}(\lambda_1)}{x_u x_v}. \tag{2.8}$$

This argument includes $h = 1$, with no interior equations. By (2.1) applied to $\mathbf{y}$,

$$\delta = \sum_{ab \in E(G)} x_a x_b \left(\frac{y_a}{x_a} - \frac{y_b}{x_b} \right)^2 = \sum_{ab \in E(G)} x_a x_b \left(\frac{w_a}{x_a} - \frac{w_b}{x_b} \right)^2.$$

Cauchy–Schwarz along the path and (2.8) give

$$\delta \geq \frac{(w_u x_v + |w_v| x_u)^2}{x_u x_v \phi_{h-1}(\lambda_1)} \geq \frac{4 w_u |w_v|}{\phi_{h-1}(\lambda_1)}.$$

□

2.3 A two-dimensional Schur-complement estimate

We record the elementary matrix argument used in both proofs.

Lemma 2.4. *Let A be the adjacency matrix of a connected graph of order at least three. Partition its vertices into two selected vertices and the remaining vertices, so that*

$$A = \begin{pmatrix} D & W^T \\ W & B \end{pmatrix}, \qquad F(z) = zI_2 - D - W^T(zI - B)^{-1}W.$$

Suppose $\lambda_2 > \rho(B)$ and $I_2 \preceq F'(z) \preceq aI_2$ for $\lambda_2 \leq z \leq \lambda_1$. Writing $F_{12}(z) = -g(z)$, one has

$$0 \leq F_{ii}(\lambda_1) \leq a\delta, \qquad \delta \geq \frac{2g(\lambda_1)}{a}.$$

Proof. Interlacing gives $\lambda_3 \leq \rho(B) < \lambda_2$, so λ_2 is simple. Block elimination is a congruence between $zI - A$ and $\operatorname{diag}(F(z), zI - B)$. Thus the two eigenvalues of $F(\lambda_2)$ are negative and zero, while those of $F(\lambda_1)$ are zero and positive. Integrating the upper derivative bound and using the variational principle gives $\lambda_{\max}(F(\lambda_1)) \leq a\delta$. The matrix $F(\lambda_1)$ is positive semidefinite and singular. Its diagonal entries are therefore nonnegative and at most $a\delta$; their product is $g(\lambda_1)^2$, so its trace is at least $2g(\lambda_1)$. Here $g(z) > 0$: a direct edge or a walk through the remaining vertices connects the two selected vertices, and the resolvent has a nonnegative walk expansion. The trace bound proves the result. Finally, differentiation gives $F'(z) = I_2 + W^T(zI - B)^{-2}W$, explaining the lower derivative bound. □

3 Two comparison families

Lemma 3.1 (Double comets)**.** *If $k \ge 3$, $L \ge 1$, and $r = (k-1)^{-1/2}$, then*

$$\delta(\mathsf{C}(k,L)) \le \frac{2(1-r^2)r^{L-1}}{1-r^{2L}} \le 2(k-1)^{-(L-1)/2}. \tag{3.1}$$

Proof. Put $\alpha = r + r^{-1} = k/\sqrt{k-1}$. Number the joining path $0, \dots, L$. Give a vector $\mathbf{u}$ the value r^j at vertex j and the value of its adjacent endpoint divided by α at each leaf. Let $\mathbf{v}$ be its reflection. Substituting into the adjacency equations gives

$$(A - \alpha I)\mathbf{u} = e\mathbf{e}_L, \quad (A - \alpha I)\mathbf{v} = e\mathbf{e}_0, \quad e = (1-r^2)r^{L-1}.$$

The positive vector $\mathbf{u} + \mathbf{v}$ implies $\lambda_1 \le \alpha + e/(1+r^L)$. The Perron vector is invariant under reflection, so $\mathbf{u} - \mathbf{v}$ is orthogonal to it. Since

$$\|\mathbf{u} - \mathbf{v}\|^2 \ge 2(1-r^L)^2, \qquad (\mathbf{u}-\mathbf{v})^T(A - \alpha I)(\mathbf{u}-\mathbf{v}) = -2e(1-r^L),$$

the variational principle gives $\lambda_2 \ge \alpha - e/(1-r^L)$. Subtracting proves the first bound. The second uses $1 - r^{2L} \ge 1 - r^2$. □

For $k \ge 3$ define

$$\beta_k = \frac{k - 2 + \sqrt{k^2-4}}{2}, \qquad \kappa_k = \beta_k + \beta_k^{-1}. \tag{3.2}$$

Lemma 3.2 (Double kites)**.** *For $k \ge 3$ and $L \ge 1$,*

$$\delta(\mathsf{K}(k,L)) \le \frac{32}{k+3}\beta_k^{-(L-1)}, \qquad \kappa_k - (k-1) > \frac{1}{k(k-1)}. \tag{3.3}$$

Proof. In the unit Perron vector of $\mathsf{K}(k,L)$, let a be the common coordinate at the two clique roots. Its maximum degree is k, so $\lambda_1 \le k$. Each nonroot clique coordinate equals $a/(\lambda_1 - k + 2) \ge a/2$; consequently $a^2 \le 2/(k+3)$.

For completeness, β_k satisfies $\beta_k^2 - (k-2)\beta_k - (k-2) = 0$. At $z = \kappa_k$ this gives

$$z - \frac{k-1}{z-k+2} = \beta_k^{-1}.$$

Use the path values r^j, where $r = \beta_k^{-1}$, and use at each nonroot clique vertex its root value divided by $\kappa_k - k + 2$. The residual of $A - \kappa_k I$ is positive only at the far root. Adding its reflection gives a positive vector with a nonzero nonnegative residual. Multiplication by the Perron vector proves $\lambda_1 > \kappa_k > 2$. Hence $r_1 = r(\lambda_1) < \beta_k^{-1}$. The exact Perron coordinates along the joining path are

$$x_j = a\frac{r_1^j + r_1^{L-j}}{1 + r_1^L} \qquad (0 \le j \le L).$$

Apply (2.2) to a function taking values $+1$ and -1 on the two reflected halves. If L is odd, only the central edge contributes and its energy is at most $16a^2 r_1^{L-1}$; the weighted norm is one. If L is even, give the central vertex value zero. The energy of the two central edges is at most $8a^2 r_1^{L-1}$. Both neighboring Perron coordinates exceed the central coordinate, because $\lambda_1 > 2$; thus the squared weighted norm is at least $2/3$. In either case the weighted mean is zero by reflection. These estimates, $a^2 \le 2/(k+3)$, and $r_1 < \beta_k^{-1}$ prove the gap bound.

Finally set $\varepsilon = k - 1 - \beta_k$. Direct substitution gives $\varepsilon^2 - k\varepsilon + 1 = 0$, so $\varepsilon > 1/k$. The same identity gives $\kappa_k - (k-1) = \varepsilon/\beta_k > 1/[k(k-1)]$. □

4 The spectral scale and concentration of branching in trees

Consider an arbitrary sequence of trees attaining τ_n, with their orders tending to infinity. In this and the following two sections, every asymptotic statement is along this sequence. We suppress the argument T in its eigenvalues, write $\delta = \tau_n$, and set

$$D = \operatorname{diam}(T), \qquad m = n - D - 1, \qquad a_n = \log n - \log\log n - \log 2.$$

Thus m is the number of vertices outside any diametral path. The arbitrariness of the sequence will give a uniform eventual conclusion.

Lemma 4.1. *Along every such sequence,*

$$-\log\delta = \frac{n}{2}(a_n - 1) + o(n), \tag{4.1}$$

$$m \sim \frac{n}{\log n}, \tag{4.2}$$

$$\log\lambda_1 = \frac{1}{2}a_n + o(1), \qquad \lambda_1^2 \sim \lambda_2^2 \sim \frac{m}{2}. \tag{4.3}$$

In particular, $n^C\delta \longrightarrow 0$ for every fixed $C > 0$.

Proof. Choose $k = \lfloor n/(2\log n)\rfloor$ and $L = n - 2k - 1$. The tree $\mathsf{C}(k, L)$ has order n, and Lemma 3.1 therefore gives

$$-\log\delta \geq \left(\frac{n}{2} - k - 1\right)\log(k-1) - \log 2 = \frac{n}{2}(a_n - 1) + o(n). \tag{4.4}$$

Indeed, $\log(k-1) = a_n + O(\log n/n)$ and $k\log(k-1) = n/2 + o(n)$. This already proves the last assertion.

Fix a diametral path P. Regard its adjacency matrix as indexed by $V(T)$, with zero rows at vertices outside the path, and write $A(T) = A(P) + A(F)$, where F consists of the edges outside P. The forest F has exactly m edges. Spectral symmetry of a bipartite graph and $\operatorname{tr} A(F)^2 = 2m$ show that

$$\sum_{\lambda_j(F)>0} \lambda_j(F)^2 = m, \qquad \lambda_2(F) \leq \sqrt{m/2}.$$

For the second assertion, there is nothing to prove if $\lambda_2(F) \leq 0$; otherwise the two largest positive squared eigenvalues are each at least $\lambda_2(F)^2$. Weyl's inequality now gives

$$\lambda_2 \leq \lambda_2(F) + \lambda_1(P) < \sqrt{m/2} + 2.$$

As $\delta < 1$ eventually, we also have $\lambda_1 < \sqrt{m/2} + 3$. Lemma 2.1 consequently implies

$$-\log\delta \leq (n - m - 2)\log\bigl(3 + \sqrt{m/2}\bigr) + O(\log n). \tag{4.5}$$

First, $m \to \infty$: if m were bounded on a subsequence, the right-hand side would be $O(n)$, contradicting (4.4). Next, $m/n \to 0$. Otherwise, on a subsequence with $m \geq \varepsilon n$ for some $\varepsilon > 0$, the coefficient of $n\log n$ in (4.5) would be at most $(1-\varepsilon)/2$, again contradicting the benchmark. Since

$$\log\bigl(3 + \sqrt{m/2}\bigr) = \frac{1}{2}\log(m/2) + O(m^{-1/2}),$$

equation (4.5) becomes

$$-\log\delta \leq \frac{n-m}{2}\log(m/2) + o(n).$$

Here the discarded error is $O(n/\sqrt{m} + \log n) = o(n)$.

Put $c = m \log n/n$. Comparison with (4.4) yields

$$0 \le \log c - c + 1 + \frac{m}{n} \log \frac{2n}{m} + o(1). \tag{4.6}$$

To see this directly, use $\log(m/2) = a_n + \log c$ and $(m/n)\log(m/2) = c - (m/n)\log(2n/m)$. The penultimate term in (4.6) tends to zero. The function $\log c - c + 1$ is nonpositive, vanishes only at $c = 1$, and tends to $-\infty$ as $c \to 0$ or $c \to \infty$. Thus $c \to 1$, proving (4.2).

Apply Lemma 2.1 once more, now retaining λ_1:

$$\log \lambda_1 \ge \frac{\frac{n}{2}(a_n - 1) + o(n)}{n - m - 2} = \frac{1}{2} a_n + o(1).$$

The last equality follows from $(a_n - 1)m/n \to 1$ and $a_n(m/n)^2 \to 0$. Conversely, $\lambda_1 < \sqrt{m/2} + 3$ and (4.2) give $\log \lambda_1 \le a_n/2 + o(1)$. It follows that $\lambda_1^2 \sim m/2$. The same holds for λ_2^2, since $\lambda_1 - \lambda_2 = \delta$ is smaller than every fixed inverse power of n. Finally,

$$-\log \delta \le (n - m - 2)\log \lambda_1 + O(\log n) = \frac{n}{2}(a_n - 1) + o(n),$$

where we used $ma_n = n(1 + o(1))$. This completes the proof. □

Lemma 4.2. *Let H be a nontrivial tree with $\lambda_1(H) > 2$, and let $\mathbf{z} > 0$ be a Perron eigenvector of H. If H has a leaves, then*

$$(\lambda_1(H) - 2) \sum_{v \in V(H)} z_v \le (a - 2) \max_{v \in V(H)} z_v.$$

Proof. Sum the eigenvalue equations and subtract twice the sum of the coordinates. The resulting identity is

$$(\lambda_1(H) - 2) \sum_v z_v = \sum_v (d_H(v) - 2) z_v.$$

Discard the negative terms, and use $\sum_v (d_H(v) - 2)_+ = a - 2$. □

Lemma 4.3. *There are distinct vertices u_1, u_2 of degrees d_1, d_2 such that*

$$d_1 \sim d_2 \sim m/2. \tag{4.7}$$

Moreover, if $B = A(T - \{u_1, u_2\})$, then

$$\|B\| = o(\sqrt{m}) = o(\lambda_2). \tag{4.8}$$

Proof. The unit eigenvector $\mathbf{y}$ corresponding to λ_2 has both signs, because it is orthogonal to the positive unit eigenvector $\mathbf{x}$ corresponding to λ_1. Choose a connected component H_1 of the subgraph induced by $\{v : y_v > 0\}$ and a connected component H_2 of the subgraph induced by $\{v : y_v < 0\}$. At a vertex of H_1, the omitted neighbors have nonpositive coordinates, so

$$A(H_1)\mathbf{y}_{H_1} \ge \lambda_2 \mathbf{y}_{H_1}.$$

The analogous inequality holds for $-\mathbf{y}_{H_2} > 0$. Multiplying by positive Perron eigenvectors of the respective components, and applying interlacing for the upper bound, gives

$$\lambda_2 \le \lambda_1(H_i) \le \lambda_1 \qquad (i = 1, 2). \tag{4.9}$$

In particular, the components are nontrivial and have spectral radius greater than two eventually.

The endpoints of a diametral path are leaves, and no internal vertex of that path is a leaf. Thus T has at most $m+2$ leaves. Every nontrivial connected subtree H has at most this many leaves, since

$$\#\{\text{leaves of } H\} - 2 = \sum_{v \in V(H)} (d_H(v) - 2)_+ \le \sum_{v \in V(T)} (d_T(v) - 2)_+ \le m.$$

Let $\mathbf{z} > 0$ be a Perron eigenvector of H_i, and choose u_i at a maximum coordinate M. By Lemma 4.2,

$$\frac{\sum_v z_v}{M} \le \frac{m}{\lambda_1(H_i) - 2} = O(\sqrt{m}).$$

There is exactly one walk of length two from u_i to each vertex at distance two in a tree, and there are $d_{H_i}(u_i)$ such walks back to u_i. Consequently,

$$\lambda_1(H_i)^2 M = d_{H_i}(u_i) M + \sum_{\mathrm{dist}_{H_i}(u_i, v) = 2} z_v.$$

Together with (4.9), this implies

$$d_i = d_T(u_i) \ge \lambda_2^2 - O(\sqrt{m}) = m/2 - o(m).$$

The vertices u_1, u_2 are distinct. Their contributions to the positive degree excess are $d_1 - 2$ and $d_2 - 2$ eventually, while the total excess is at most m. Hence both degrees are asymptotic to $m/2$, and the total positive degree excess at all remaining vertices is $o(m)$. Their degrees are therefore uniformly $o(m)$.

For completeness, a forest of maximum degree $\Delta \ge 1$ has operator norm at most $2\sqrt{\Delta}$. Orient each component away from a root. For an edge with parent a and child b, use

$$2|z_a z_b| \le \Delta^{-1/2} z_a^2 + \Delta^{1/2} z_b^2.$$

Each vertex is a parent at most Δ times and a child at most once, so summation bounds the absolute adjacency quadratic form by $2\sqrt{\Delta}\|\mathbf{z}\|^2$. An edgeless forest has norm zero. Apply this estimate to $T - \{u_1, u_2\}$, whose maximum degree is $o(m)$, to obtain (4.8). □

5 Equal degrees at the two dominant vertices

Retain u_1, u_2 and B from Lemma 4.3. Let $\mathbf{a}_i$ be the indicator vector of the neighbors of u_i in $T - \{u_1, u_2\}$, let $W = [\mathbf{a}_1\ \mathbf{a}_2]$, and let $A_U = A(T[\{u_1, u_2\}])$. For $z > \|B\|$, define

$$S(z) = W^{\mathsf{T}}(zI - B)^{-1} W, \qquad F(z) = zI_2 - A_U - S(z). \tag{5.1}$$

We write $F_{12}(z) = -g(z)$; explicitly,

$$g(z) = \mathbf{1}_{\{u_1 u_2 \in E(T)\}} + \mathbf{a}_1^{\mathsf{T}}(zI - B)^{-1}\mathbf{a}_2 > 0.$$

The strict inequality follows from the unique path between the two vertices and the nonnegative Neumann expansion of the resolvent.

Lemma 5.1. *Eventually, throughout $\lambda_2 \le z \le \lambda_1$,*

$$I_2 \preceq F'(z) \preceq 4I_2.$$

Consequently,

$$\delta \ge \frac{1}{2} g(\lambda_1), \qquad 0 \le F_{ii}(\lambda_1) \le 4\delta \quad (i = 1, 2). \tag{5.2}$$

Proof. By (4.8), $\lambda_2 > \|B\|$ eventually. Interlacing gives $\lambda_3 \le \lambda_1(B) < \lambda_2$, where $\lambda_1(B)$ denotes the largest eigenvalue of the matrix B. Thus the two-root Schur complement in Lemma 2.4 applies. Direct differentiation gives

$$F'(z) = I_2 + W^{\mathsf{T}}(zI - B)^{-2}W.$$

The second term is positive semidefinite, and its norm is at most

$$\frac{\|W\|^2}{(z - \|B\|)^2} \le \frac{d_1 + d_2}{(\lambda_2 - \|B\|)^2} = 2 + o(1).$$

The upper bound $4I_2$ follows eventually. Apply Lemma 2.4 with derivative bound four to obtain (5.2). □

Lemma 5.2. *Let $L = \mathrm{dist}_T(u_1, u_2)$. Then*

$$L \ge n - (1 + o(1))\frac{n}{\log n}. \tag{5.3}$$

In particular, $L \ge 3$ eventually. The integer

$$b = n - (L + 1) - (d_1 - 1) - (d_2 - 1) \tag{5.4}$$

is nonnegative and satisfies $b = o(m)$.

Proof. For $L \ge 2$, the unique connecting path contributes the term $z^{-(L-1)}$ to the Neumann expansion of $g(z)$. For $L = 1$ the direct edge gives $g(z) \ge 1$, so in all cases $g(\lambda_1) \ge \lambda_1^{-(L-1)}$. By (5.2),

$$(L - 1)\log\lambda_1 \ge -\log\delta - \log 2.$$

Equations (4.4) and (4.3) give

$$L - 1 \ge n\frac{a_n - 1 + o(1)}{a_n + o(1)} = n - (1 + o(1))\frac{n}{\log n}.$$

The additive $o(1)$ precision in $\log\lambda_1$ is used here.

The path has $L+1$ vertices. At its endpoints there are $d_1 - 1$ and $d_2 - 1$ neighbors outside the path. These neighbors are all distinct, by acyclicity, so $b \ge 0$. Finally, (4.2), (4.7), and (5.3) yield

$$0 \le b = n - L + 1 - d_1 - d_2 = o(n/\log n) = o(m).$$

□

Lemma 5.3. *For $L \ge 3$ and $z > \|B\|$,*

$$S_{ii}(z) = \frac{d_i}{z} + E_i(z), \qquad 0 \le E_i(z) \le \frac{b + 1}{z(z^2 - \|B\|^2)}. \tag{5.5}$$

Proof. Since the hubs are not adjacent, $\|\mathbf{a}_i\|^2 = d_i$. Each entry of $B\mathbf{a}_i$ is zero or one: two neighbors in the support of $\mathbf{a}_i$ would create a cycle of length four through u_i. Its nonzero entries therefore count vertices at distance two from u_i reached without passing through the other hub. Exactly one such vertex lies on the connecting path. Any other such vertex lies outside that path and cannot be an immediate off-path neighbor of either hub. For its own hub this would create a triangle; for the other hub it would create a second path between the hubs and hence a cycle. Thus every such vertex is among the b vertices counted by (5.4), and

$$\|B\mathbf{a}_i\|^2 \le b + 1. \tag{5.6}$$

All neighbors of u_i lie in the same bipartition class of T. Consequently $\mathbf{a}_i^{\mathsf{T}} B^{2j+1} \mathbf{a}_i = 0$ for every $j \geq 0$. Expanding the resolvent gives

$$S_{ii}(z) = \frac{d_i}{z} + \sum_{j \geq 1} \frac{\|B^j \mathbf{a}_i\|^2}{z^{2j+1}}.$$

For $j \geq 1$, the numerator is at most $\|B\|^{2j-2} \|B\mathbf{a}_i\|^2$. Summing the resulting geometric series proves the claim. □

Lemma 5.4. *Eventually $d_1 = d_2 = k+1$ for an integer k, and*

$$k \sim \frac{n}{2 \log n}, \qquad b = o(k), \qquad n = 2k + L + 1 + b. \tag{5.7}$$

Moreover,

$$\lambda_1^2 \leq k + 1 + \frac{3(b+1)}{k}. \tag{5.8}$$

Proof. From the definitions, there is the exact identity

$$\lambda_1^2 - d_i = \lambda_1 E_i(\lambda_1) + \lambda_1 F_{ii}(\lambda_1).$$

Use (5.5) and (5.2) to obtain

$$d_i \leq \lambda_1^2 \leq d_i + \frac{b+1}{\lambda_1^2 - \|B\|^2} + 4\lambda_1 \delta. \tag{5.9}$$

The error on the right tends to zero: $b = o(m)$, $\lambda_1^2 - \|B\|^2 \sim m/2$, and $\lambda_1 \delta \to 0$. The integers d_1, d_2 therefore lie in the same interval of length $o(1)$ and must be equal eventually. Their common value and the definition of b give (5.7).

For large n, $\lambda_1^2 - \|B\|^2 \geq k/2$. Also $4\lambda_1 \delta \leq (b+1)/k$, because $b + 1 \geq 1$, $k\lambda_1$ grows at most polynomially in n, and δ is smaller than every fixed inverse power of n. Substitution in (5.9) proves (5.8). □

6 Exact completion of the tree structure

Lemma 6.1. *Write $\lambda_1 = r + r^{-1}$ with $0 < r < 1$, and put $t = r^{-2}$. Along the minimizing sequence, eventually*

$$\delta \geq \frac{1}{4} t^{-(L-1)/2}, \qquad t \leq k - 1 + \frac{3(b+1)}{k}. \tag{6.1}$$

Proof. The representation of λ_1 is valid eventually because $\lambda_1 \to \infty$. Retain in the residual resolvent only those walks confined to the interior of the path between u_1 and u_2. All terms are nonnegative, so Lemma 2.2 gives

$$g(\lambda_1) \geq \left((\lambda_1 I - A(P_{L-1}))^{-1} \right)_{1,L-1} = \frac{1}{\phi_{L-1}(\lambda_1)} = \frac{(1-r^2) r^{L-1}}{1 - r^{2L}}.$$

Since $r \to 0$, the last expression is at least $r^{L-1}/2$ eventually. The first assertion now follows from $\delta \geq g(\lambda_1)/2$. For the second, use $\lambda_1^2 = t + 2 + t^{-1}$ and (5.8), discarding the positive term t^{-1}. □

Proof of Theorem 1.1. Suppose that there are minimizers which are not symmetric double comets at arbitrarily large orders. Choose a sequence of these minimizers and apply the preceding lemmas. The symmetric double comet

$$C = \mathsf{C}(k, L + b)$$

has the same order $n = 2k + L + 1 + b$. By Lemmas 3.1 and 6.1,

$$\frac{\delta(T)}{\delta(C)} \geq \frac{1}{8}(k-1)^{b/2}\left(1 + \frac{3(b+1)}{k(k-1)}\right)^{-(L-1)/2}. \tag{6.2}$$

Taking logarithms and using $\log(1+s) \leq s$ for $s \geq 0$ yields

$$\log\frac{\delta(T)}{\delta(C)} \geq \frac{b}{2}\log(k-1) - \frac{3(L-1)(b+1)}{2k(k-1)} - \log 8. \tag{6.3}$$

By (5.7), $L \leq n = O(k \log k)$. If $b \geq 1$, then $b + 1 \leq 2b$, and the right-hand side is bounded below by

$$b\left(\frac{1}{2}\log(k-1) - O\left(\frac{\log k}{k}\right)\right) - \log 8.$$

This tends to $+\infty$ uniformly over integers $b \geq 1$, contradicting the minimality of T.

It follows that $b = 0$ eventually. Every vertex then lies on the connecting path or is an immediate off-path neighbor of one of its endpoints. Acyclicity forces each such off-path neighbor to be a leaf: the path together with its incident edges to these neighbors already forms a spanning tree. The equality $d_1 = d_2 = k + 1$ gives exactly k leaves at each endpoint. Hence $T = \mathsf{C}(k, L)$, contradicting our choice of the sequence. This proves the existence of a uniform eventual threshold. Finally, $k \sim n/(2\log n)$ and $L = n - 2k - 1$ give the stated parameter asymptotics. □

7 The global scale and vertex budget for graphs

We now consider an arbitrary sequence $G \in \mathcal{G}_n$ with $\delta(G) = \eta_n$ and $n \to \infty$. Use the centered vector $\mathbf{w} = \mathbf{y} - c\mathbf{x}$ of Lemma 2.3, and choose vertices a, b with $w_a = M = -w_b$. Write $d = \operatorname{dist}(a, b)$ and

$$k_0 = \lceil \lambda_2 - \sqrt{n}\,\delta \rceil. \tag{7.1}$$

The following counting bound is valid before imposing minimality, whenever $k_0 \geq 1$.

Lemma 7.1. *One has $d \leq n - 2k_0 + 1$. If also $\lambda_1 > 2$, then*

$$\delta \geq \frac{4}{n}\lambda_1^{-n+2\lambda_1-2(1+\sqrt{n})\delta}. \tag{7.2}$$

Proof. At the positive extremum, (2.6) gives

$$\lambda_2 M = \sum_{v\sim a} w_v + c\delta x_a \leq M|N(a) \cap \{w > 0\}| + |c|\delta x_a.$$

Since $x_a \leq 1$ and $|c|/M \leq \sqrt{n}$, at least k_0 neighbors of a have positive centered coordinates. The negative extremum has at least k_0 negative neighbors. These two sets are disjoint and each meets a shortest a–b path in at most one vertex. Their union with that path has at least $d + 1 + 2k_0 - 2$ vertices. This proves the distance bound and, by (7.1),

$$d - 1 \leq n - 2\lambda_1 + 2(1 + \sqrt{n})\delta.$$

Lemma 2.3 at a, b gives $\delta \geq 4M^2/\phi_{d-1}(\lambda_1)$. Now use $M^2 \geq 1/n$ and $\phi_{d-1}(\lambda_1) \leq \lambda_1^{d-1}$. □

Lemma 7.2. *Let $z_n > 1$ be the solution of $z_n(1+\log z_n) = n/2$, defined for all sufficiently large n. Then*

$$-\log \eta_n = (n-2z_n)\log z_n + O(\log n), \tag{7.3}$$

$$\lambda_1 = z_n + O\left(\sqrt{n/\log n}\right), \qquad k_0 \sim \frac{n}{2\log n}. \tag{7.4}$$

In particular, δ is smaller than every fixed inverse power of n eventually.

Proof. The function $f_n(z) = (n-2z)\log z$, $z \geq 1$, has a unique maximum at z_n. Take an integer h nearest to $z_n + 2$, and compare with $\mathsf{K}(h, n-2h+1)$. Since $\beta_h > h-2$, Lemma 3.2 gives

$$-\log \eta_n \geq (n-2h)\log(h-2) - O(\log n) \geq f_n(z_n) - O(\log n).$$

To justify the second estimate, put $u = h-2 = z_n + O(1)$. The expression on the left is $f_n(u) - 4\log u$; Taylor's theorem at z_n gives $f_n(u) = f_n(z_n) + O((\log n)^2/n)$. This lower bound proves the asserted rapid decay of δ.

Also $\lambda_1 \to \infty$. Otherwise, along a subsequence on which it is bounded, Lemma 2.1, with $D \leq n-1$, would give $-\log\delta = O(n + \log n)$, a contradiction. Thus $\lambda_1 > 2$ and $k_0 \geq 1$ eventually. Applying (7.2), and using $\lambda_1 \leq n-1$ and the rapid decay of δ, gives

$$-\log \eta_n \leq f_n(\lambda_1) + O(\log n) \leq f_n(z_n) + O(\log n).$$

This proves (7.3). If $t = \lambda_1/z_n$, direct algebra gives

$$f_n(z_n) - f_n(tz_n) = 2z_n\big[\log z_n(t - 1 - \log t) + (t-1)\log t\big].$$

The two bracketed terms are nonnegative. Since their sum multiplied by $2z_n$ is $O(\log n)$, first $t \to 1$, and then $(t-1)^2 = O(\log n/n)$. Because $z_n \sim n/(2\log n)$, the claimed bound on λ_1 follows. The definition of k_0 changes λ_1 by only $O(1)$, proving its scale. □

Lemma 7.3 (A constant vertex budget)**.** *Eventually the nonnegative integer*

$$b_0 = n - d - 2k_0 + 1$$

belongs to $\{0, 1, 2\}$.

Proof. Nonnegativity follows from Lemma 7.1. Put $h = k_0 + 1$, $B_0 = \beta_h$, and $m = d - 1 = n - 2k_0 - b_0$. The definition of k_0 and (3.3) give

$$\lambda_1 \leq k_0 + (1+\sqrt{n})\delta < \kappa_h$$

eventually; the separation of κ_h from k_0 is at least $1/[k_0(k_0+1)]$, whereas the error is superpolynomially small. For $\gamma = (\lambda_1 + \sqrt{\lambda_1^2 - 4})/2$, eventually $2 \leq \gamma < B_0$. Lemma 2.2 implies

$$\phi_m(\lambda_1) = \frac{\gamma^{m+1} - \gamma^{-m-1}}{\gamma - \gamma^{-1}} \leq \frac{4}{3}B_0^m.$$

The exact resistance bound at the extrema yields $\delta \geq 3B_0^{-m}/n$. On the other hand, the same-order competitor $\mathsf{K}(k_0+1, n-2k_0-1)$ gives

$$\delta \leq \frac{32}{k_0+4}B_0^{-(n-2k_0-2)}.$$

Consequently $B_0^{b_0-2} \leq 32n/[3(k_0+4)]$. If $b_0 \geq 3$, this forces $B_0(k_0+4) \leq 32n/3$, contrary to $B_0 \sim k_0 \sim n/(2\log n)$. □

8 Dense terminal cores in a minimizing graph

Fix $\theta = 0.99$ and a shortest path $Q = (a = v_0, \dots, v_d = b)$. Define

$$A_+ = N(a) \cap \{w > 0\}, \qquad B_- = N(b) \cap \{w < 0\}, \qquad E = V(G) \setminus (V(Q) \cup A_+ \cup B_-).$$

Let I_A be the indicator of $v_1 \in A_+$ and I_B that of $v_{d-1} \in B_-$. Inclusion–exclusion gives the exact identity

$$b_0 = (|A_+| - k_0) + (|B_-| - k_0) + (2 - I_A - I_B) + |E|. \tag{8.1}$$

Each term is nonnegative. Thus $|A_+|, |B_-| \le k_0 + 2$ and $|E| \le 2$. We use several consequences of shortestness. There is no edge from A_+ to B_-, since such an edge would give $d \le 3$. A vertex of A_+ can have path neighbors only at positions $0, 1, 2$; the reversed assertion holds for B_-. In general, the path neighbors of any vertex outside a geodesic have indices spanning at most two: otherwise the two incident edges would shorten the geodesic. In particular there are at most three such neighbors.

Lemma 8.1. *Let*

$$C_1 = \{v : w_v \ge \theta M\}, \qquad C_2 = \{v : w_v \le -\theta M\}.$$

Then, along the minimizing sequence:

1. $|C_i| = k_0 + O(1)$*, and the minimum degree of* $G[C_i]$ *is at least* $0.9|C_i| - 1$ *eventually;*
2. *every vertex of* C_i *is within distance two of its corresponding extremum, and*

$$\sum_{v \in C_1} (1 - w_v/M) = O(1), \qquad \sum_{v \in C_2} (1 + w_v/M) = O(1); \tag{8.2}$$

3. *the residual graph* $G - (C_1 \cup C_2)$ *has bounded maximum degree and bounded adjacency norm. Only* $O(1)$ *residual vertices meet the cores, and every residual coordinate has absolute value less than* θM*.*

Proof. Discard the negative neighbor contributions in the equation at a. This gives

$$\sum_{v \in A_+} (1 - w_v/M) \le |A_+| - \lambda_2 + \sqrt{n}\,\delta \le 3 + o(1). \tag{8.3}$$

Since $1 - \theta$ is fixed, only $O(1)$ vertices of A_+ fail the threshold defining C_1. The analogous assertion holds at b.

For $3 \le j \le d - 3$, the vertex v_j has at most its two path neighbors and the vertices of E as neighbors. Hence its degree is at most four, and (2.6) yields

$$|w_{v_j}| \le \frac{4 + \sqrt{n}\,\delta}{\lambda_2} M < \theta M.$$

A high positive path vertex cannot be among the last three vertices of Q: its positive neighbors would then be confined to a bounded set. Similarly a high negative vertex cannot be at the positive end. A high positive vertex in E needs at least $\theta\lambda_2 - o(1)$ positive neighbors. Outside A_+ it has at most three path neighbors and one other neighbor in E, and no positive neighbor in B_-. Thus it meets A_+ and lies within distance two of a. Reverse signs for the assertion about C_2.

It follows that C_1 consists of the high vertices of A_+, the extremum a, and at most a bounded number of further path or E vertices. This proves its order estimate, its localization, and its deficit bound from (8.3). The same holds for C_2. Every vertex of C_1 needs $\theta\lambda_2 - o(1) =$

$\theta k_0 - O(1)$ positive neighbors. All but $O(1)$ of these lie in C_1: the remaining possibilities are the omitted vertices of A_+, a bounded number of path vertices, and E. The opposite terminal set is too far away to contribute. Hence its internal degree is at least $\theta k_0 - O(1)$, which is at least $0.9|C_1| - 1$ eventually. The other core is identical in this respect.

After both cores are removed, only $O(1)$ vertices outside Q remain. Each has at most three neighbors on Q, so the residual graph has bounded maximum degree and therefore bounded norm. The path neighbors of a core vertex are confined to the first or last four positions, because that core vertex is within distance two of the corresponding extremum. Thus only boundedly many residual vertices meet either core. Finally, the cores were defined using *all* coordinates at the threshold, so every residual coordinate is strictly below it in absolute value. □

9 Transmission and attachment estimates

Interchange the two ends if necessary so that $|C_1| = k$ and $|C_2| = k + q$, where $q \geq 0$. When doing so replace $(\mathbf{w}, \mathbf{y}, c)$ by $(-\mathbf{w}, -\mathbf{y}, -c)$, leaving $\mathbf{x}$ unchanged. Thus $\mathbf{w}|_{C_1} > 0$ and $\mathbf{w}|_{C_2} < 0$. By Lemma 8.1, $q = O(1)$, $k = k_0 + O(1)$, and $\lambda_1, \lambda_2 = k + O(1)$.

Let $\mathcal{R} = V(G) \setminus (C_1 \cup C_2)$, $A_i = A(G[C_i])$, and $B = A(G[\mathcal{R}])$. Let W_i be the matrix with rows indexed by C_i and columns by $\mathcal{R}$ that records the edges between these sets. Only boundedly many columns are nonzero; consequently $\|W_i\|^2 = O(k)$. Write $\mathbf{w}_i, \mathbf{x}_i$ for restrictions to C_i, and use subscript $\mathcal{R}$ for restrictions to the residual set. Put $R_2 = (\lambda_2 I - B)^{-1}$, which is entrywise nonnegative and has norm $O(1/k)$.

Choose a shortest path between C_1 and C_2, of L edges. Its first and last internal vertices are s_1, s_2; let $d_i \geq 1$ count the neighbors of s_i in its adjacent core. Define

$$b = n - |C_1| - |C_2| - (L - 1) \geq 0. \tag{9.1}$$

Core localization gives $d - 4 \leq L \leq d$. Therefore

$$b = O(1), \qquad L = n - 2k + O(1), \qquad \frac{L}{k \log k} \longrightarrow 2. \tag{9.2}$$

No internal path vertex other than s_1, s_2 meets a core, since that would shorten the path. No residual vertex meets both cores. Both assertions hold eventually because $L \to \infty$.

Lemma 9.1 (Endpoint transmission)**.** *Along every minimizing sequence,*

$$\delta \geq \frac{2}{k}(\theta^2 - o(1)) d_1 d_2 \lambda_1^{-(L-1)}. \tag{9.3}$$

Proof. The residual part of (2.6) is exactly

$$\mathbf{w}_{\mathcal{R}} = R_2(W_1^T \mathbf{w}_1 + W_2^T \mathbf{w}_2 + c\delta \mathbf{x}_{\mathcal{R}}). \tag{9.4}$$

The source from the cores has bounded support and coordinates of magnitude $O(kM)$, so its norm is $O(kM)$. The forcing is negligible because $|c|/M \leq \sqrt{n}$ and δ is superpolynomially small. It follows that $\|\mathbf{w}_{\mathcal{R}}\| = O(M)$. Thus

$$1 + c^2 = \|\mathbf{w}\|^2 \leq (2k + O(1))M^2, \qquad M^2 \geq \frac{1 - o(1)}{2k}.$$

The zeroth term of the Neumann series for R_2 contributes at least $\theta d_1 M/\lambda_2$ to w_{s_1} from the positive core. The opposite-core source has residual distance at least $L - 2$ from s_1. If there

were a shorter residual path, adjoining one edge into each core would contradict the choice of L. Its total contribution has absolute value at most

$$O(kM)\frac{\|B\|^{L-2}}{\lambda_2^{L-1}(1-\|B\|/\lambda_2)}.$$

This is smaller than M times every fixed inverse power of k, since $\|B\| = O(1)$ and $L \sim 2k\log k$. The forcing term in (9.4) has the same property. Therefore

$$w_{s_1} \geq (\theta - o(1))\frac{d_1 M}{\lambda_2}, \qquad -w_{s_2} \geq (\theta - o(1))\frac{d_2 M}{\lambda_2}.$$

In the full graph the distance between s_1, s_2 is exactly $L-2$; a shorter path would again yield a shorter walk between the cores. Apply (2.7), use $\phi_{L-3}(\lambda_1) \leq \lambda_1^{L-3}$, and discard the factor $\lambda_1^2/\lambda_2^2 \geq 1$. The resulting bound is (9.3). □

For $v \in \mathcal{R}$ let $d_i(v) = |N(v) \cap C_i|$, and define

$$\sigma_i = \sum_{v\in\mathcal{R}} \left(\frac{d_i(v)}{k}\right)^2. \tag{9.5}$$

Only boundedly many terms are nonzero.

Lemma 9.2 (A finite-rank spectral bound)**.** *Uniformly over residual vertices,*

$$d_i(v)/k \leq \theta + O(1/k). \tag{9.6}$$

Moreover,

$$\lambda_1 \leq k - 1 + \min(\sigma_1, q + \sigma_2) + O(1/k). \tag{9.7}$$

Proof. Each residual vertex has bounded residual degree and meets at most one core. By the *total* deficit estimate (8.2), the sum of absolute centered coordinates of its neighbors in that core is at least $d_i(v)M - O(M)$. Its residual neighbors contribute only $O(M)$ in absolute value. Hence the approximate eigenvector equation implies

$$d_i(v)M - O(M) \leq \lambda_2|w_v| + |c|\delta x_v \leq \theta\lambda_2 M + o(M).$$

Since $\lambda_2 = k + O(1)$, this proves (9.6) even if w_v has the sign opposite to that core.

Put $S_{ij} = W_i R_2 W_j^T$. Substitution of (9.4) in the core equations gives

$$(A_i + S_{ii})\mathbf{w}_i = \lambda_2\mathbf{w}_i - S_{ij}\mathbf{w}_j - c\delta(\mathbf{x}_i + W_i R_2 \mathbf{x}_{\mathcal{R}}), \qquad j \neq i. \tag{9.8}$$

The cross term has a favorable sign: it is nonnegative in the positive core equation, and becomes nonnegative after multiplying the negative core equation by -1. Multiply by a positive unit Perron vector of $A_i + S_{ii}$. The denominator from $|\mathbf{w}_i|$ is at least θM times the sum of its coordinates. The vector in parentheses has uniformly bounded coordinates, as $\|W_i R_2\| = O(k^{-1/2})$. Thus the forcing changes the resulting quotient by at most $O(\sqrt{n}\,\delta) = o(1/k)$, and

$$\lambda_2 \leq \rho(A_i + S_{ii}) + o(1/k). \tag{9.9}$$

The dense core is connected, so the positive Perron vector used here exists; adding $S_{ii} \geq 0$ preserves irreducibility.

The resolvent expansion and $\|W_i\|^2 = O(k)$ give

$$S_{ii} = \lambda_2^{-1} W_i W_i^T + O_{\mathrm{op}}(1/k), \qquad \|S_{ii}\| = O(1).$$

Write $a_i = |C_i|$ and let $\mathbf{u}$ be the uniform unit vector on a_i coordinates. Since $A_i \le J - I$ entrywise, Perron monotonicity and the spectral separation of the uniform direction yield

$$\rho(A_i + S_{ii}) \le \rho(J - I + S_{ii}) \le a_i - 1 + \mathbf{u}^T S_{ii}\mathbf{u} + O(1/k). \tag{9.10}$$

Here is an explicit justification of the last error. For a unit vector $t\mathbf{u}+\mathbf{v}$ with $\mathbf{v} \perp \mathbf{u}$, subtract $a_i - 1 + \mathbf{u}^T S_{ii}\mathbf{u}$ from its quadratic form. If $K = \|S_{ii}\| = O(1)$, the difference is at most $-(a_i - 2K)\|\mathbf{v}\|^2 + 2K\|\mathbf{v}\|$, whose maximum is $K^2/(a_i - 2K) = O(1/k)$. Finally,

$$\mathbf{u}^T S_{ii}\mathbf{u} = \frac{\sum_v d_i(v)^2}{a_i \lambda_2} + O(1/k) = \sigma_i + O(1/k),$$

because $a_i, \lambda_2 = k + O(1)$ and $\sigma_i = O(1)$. Combine (9.9)–(9.10) for both cores and use $\lambda_1 - \lambda_2 = \delta$ to obtain (9.7). □

10 Completing two dense terminal graphs

For disjoint rooted graphs (H_1, a_1) and (H_2, a_2), denote by $\mathcal{D}(H_1, H_2; L)$ the graph obtained by joining the roots by a path of L edges, all of whose internal vertices are new. Thus $\mathcal{D}(K_k, K_k; L) = \mathsf{K}(k, L)$, and both graphs have $2k+L-1$ vertices when $|V(H_1)| = |V(H_2)| = k$. We prove a comparison that does not require the two rooted graphs to be isomorphic.

Proposition 10.1. *Suppose that $k \ge 1000$, $L \ge 10000k$, and H_1, H_2 have order k and minimum degree at least $k - 1 - k/10$. If at least one of these graphs is incomplete, then*

$$\delta\big(\mathcal{D}(H_1, H_2; L)\big) > \delta\big(\mathsf{K}(k, L)\big).$$

For a graph H rooted at a, put $B_H = A(H - a)$ and let $\mathbf{a}_H$ be its root-incidence vector: its coordinate at $v \in V(H - a)$ equals one if $av \in E(H)$ and zero otherwise. Define the inverse rooted response by

$$F_H(z) = z - \mathbf{a}_H^T (zI - B_H)^{-1}\mathbf{a}_H, \qquad z > \rho(B_H). \tag{10.1}$$

Here and below ρ denotes the spectral radius of a nonnegative symmetric matrix. Direct differentiation gives

$$\begin{aligned} F_H'(z) &= 1 + \mathbf{a}_H^T (zI - B_H)^{-2}\mathbf{a}_H, \\ F_H''(z) &= -2\mathbf{a}_H^T (zI - B_H)^{-3}\mathbf{a}_H. \end{aligned} \tag{10.2}$$

We use the path functions $r(z)$, $s_L(z)$, and $h_L(z)$ from Lemma 2.2; in particular,

$$s_L(z) - r(z) = -h_L(z) r(z)^L, \qquad h_L(z) = \frac{(1 - r(z)^2) r(z)^{L-1}}{1 - r(z)^{2L}}. \tag{10.3}$$

Lemma 10.2. *Let H have order $k \ge 1000$, minimum degree at least $k - 1 - k/10$, and M missing edges. There is a unique number $\alpha_H > \rho(H)$ such that $F_H(\alpha_H) = r(\alpha_H)$. Write α_K for this parameter when $H = K_k$. By the rooted clique equation in Lemma 3.2 and uniqueness, $\alpha_K = \kappa_k$ and $r(\alpha_K)^{-1} = \beta_k$. Then*

$$0 < \alpha_H - \rho(H) \le r(\alpha_H) \le \frac{2}{k}, \qquad \rho(H) - \rho(B_H) \ge \frac{1}{8}, \tag{10.4}$$

$$\alpha_K - \alpha_H \ge \frac{M}{4k}, \tag{10.5}$$

$$F_H'(\alpha_H) \le 4k, \qquad F_H'(\alpha_H) \le k + 12M + 6. \tag{10.6}$$

For every $z \ge \alpha_H - 1/16$, the response is defined and satisfies

$$|F_H''(z)| \le 32\big(F_H'(z) - 1\big) \le 32 F_H'(z). \tag{10.7}$$

Proof. We first establish estimates that remain valid as missing edges are inserted with weights between zero and one. Write the resulting weighted adjacency matrix as $A = J - I - E$, where E is symmetric and nonnegative and every row sum of E is at most $k/10$. For $0 \le t \le 2/k$, let ξ and $\boldsymbol{\varphi}$ be the Perron root and positive unit Perron vector of $A + t\mathbf{e}_a\mathbf{e}_a^T$. These matrices are irreducible: their underlying graphs have minimum degree greater than $(k-1)/2$. Row sums give

$$\frac{9}{10}k - 1 \le \xi \le k - 1 + \frac{2}{k}.$$

Put $T = \xi + 1$ and $S = \sum_j \varphi_j$. The coordinate equations imply

$$\varphi_{\max} \le \frac{S}{T - t}, \qquad \varphi_{\min} \ge \frac{S - (k/10)\varphi_{\max}}{T}.$$

Consequently

$$\frac{\varphi_{\min}}{\varphi_{\max}} \ge 1 - \frac{k/10 + t}{T} > \frac{3}{4}, \qquad \frac{1}{2\sqrt{k}} \le \varphi_j \le \frac{2}{\sqrt{k}}. \tag{10.8}$$

The same calculation on the root-deleted graph uses $T \ge (9/10)k - 1$ and gives these latter coordinate bounds for its unit Perron vector $\mathbf{v}$ as well. During the interpolation, B_H and $\mathbf{a}_H$ denote the current weighted nonroot block and root-incidence vector. The nonroot coordinate equation yields

$$(\xi I - B_H)\boldsymbol{\varphi}_{H-a} = \mathbf{a}_H\varphi_a.$$

Multiplication by $\mathbf{v}^T$, followed by Cauchy–Schwarz in the denominator, gives

$$\xi - \rho(B_H) = \frac{(\mathbf{v}^T\mathbf{a}_H)\varphi_a}{\mathbf{v}^T\boldsymbol{\varphi}_{H-a}} \ge \frac{(9/10)k - 1}{4k} > \frac{1}{8}.$$

Taking $t = 0$ proves the root-deletion estimate in (10.4).

The Perron equation for H shows that $F_H(\rho(H)) = 0$. Since $F_H' > 0$ and $r' < 0$, the function $F_H - r$ is strictly increasing on $[\rho(H), \infty)$, is negative at the left endpoint, and tends to infinity. Its unique zero is α_H. Eliminating the nonroot coordinates also proves the equivalent identity

$$\alpha_H = \rho(A(H) + r(\alpha_H)\mathbf{e}_a\mathbf{e}_a^T). \tag{10.9}$$

Indeed the eliminated vector is positive, so its eigenvalue is the Perron root. The variational principle now gives $0 < \alpha_H - \rho(H) \le r(\alpha_H)$. Since $\alpha_H \ge (9/10)k - 1$, we have $r(\alpha_H) \le 2/k$. Thus the preceding coordinate estimates apply to the unit Perron vector in (10.9).

Insert all M missing edges with a common weight $s \in [0, 1]$, and denote the parameter and unit Perron vector by α_s and $\boldsymbol{\varphi}(s)$. The eigenvalue is simple, and differentiating (10.9) gives

$$\alpha_s' = \frac{2\sum_{uv \text{ missing}} \varphi_u(s)\varphi_v(s)}{1 - r'(\alpha_s)\varphi_a(s)^2}.$$

The numerator is at least $M/(2k)$. Moreover, $|r'(\alpha_s)| \le 4/k^2$ and $\varphi_a(s)^2 \le 4/k$, so the denominator is at most two. Integration proves (10.5).

The eigenvector in (10.9) with root coordinate one is

$$\left(1, (\alpha_H I - B_H)^{-1}\mathbf{a}_H\right)^T.$$

Its squared norm is $F_H'(\alpha_H)$, whence $F_H'(\alpha_H) = 1/\varphi_a^2 \le 4k$. For the sharper estimate, first assume $M \le k$ and put $\alpha = \alpha_H$. The average degree and Perron monotonicity give

$$k - 1 - \frac{2M}{k} \le \alpha \le \alpha_K < k.$$

The last strict inequality follows from the maximum row sum $k-1+r(\alpha_K)<k$ in (10.9). For the clique,

$$r(\alpha_K)^{-1}=\frac{k-1}{\alpha_K-k+2}<k-1.$$

Since $r(z)^{-1}$ is increasing, $\alpha+1-r(\alpha)=1+r(\alpha)^{-1}<k$. The root coordinate equation and $\varphi_{\max}/\varphi_{\min}\le 2$ imply

$$\frac{S}{\varphi_a}=\alpha+1-r(\alpha)+\sum_{j:\,aj\ \text{missing}}\frac{\varphi_j}{\varphi_a}\le k+2M.$$

As $1=\sum_j\varphi_j^2\le\varphi_{\max}S$ and $\varphi_{\max}\le S/(\alpha+1-r(\alpha))$, it follows that

$$F_H'(\alpha)\le\frac{(k+2M)^2}{k-2M/k-2/k}\le k+12M+6.$$

The denominator is positive. After clearing it, the last inequality is equivalent to positivity of

$$8kM-4M^2-2M+6k-2-\frac{24M^2+36M+12}{k}.$$

For $0\le M\le k$ and $k\ge 1000$, this expression is at least $(4k-27)M+6k-3>0$. If $M\ge k$, the already established bound $F_H'(\alpha_H)\le 4k$ is stronger than the claimed estimate. Finally, $\alpha_H-\rho(B_H)\ge 1/8$, so for $z\ge\alpha_H-1/16$ the resolvent $(zI-B_H)^{-1}$ is positive definite with norm at most 16. Applying this to (10.2) proves (10.7). □

Lemma 10.3. *For $k\ge 1000$ and $L\ge k$,*

$$\frac{1}{k^2}\le\alpha_K-(k-1)\le\frac{2}{k^2},\tag{10.10}$$

and

$$\delta(\mathsf{K}(k,L))\le\frac{2h_L(\alpha_K)}{k-2}.\tag{10.11}$$

In addition,

$$\delta(\mathsf{K}(k,L))\le 2h_L(\alpha_K)\le 2(k-2)^{-(L-1)}.\tag{10.12}$$

Proof. Write $\alpha=\alpha_K$, $r=r(\alpha)$, $h=h_L(\alpha)$, $\varepsilon=\alpha-(k-1)$, and $\lambda_j^K=\lambda_j(\mathsf{K}(k,L))$. The clique response is

$$F_{K_k}(z)=z-\frac{k-1}{z-k+2},$$

so $F_{K_k}(\alpha)=r$ is equivalent to

$$\varepsilon(k+\varepsilon)=r(1+\varepsilon).\tag{10.13}$$

We have $0<\varepsilon\le r\le 2/k$, and $1/k<r\le 1/(k-2)$ because $k-1<\alpha<k$. Thus (10.13) gives $\varepsilon>1/k^2$ and

$$\varepsilon\le\frac{1+2/k}{k(k-2)}\le\frac{2}{k^2},$$

proving (10.10).

Label the joining path $a_0,\dots,a_L$. Give a vector $\mathbf{u}$ the coordinates $u_{a_j}=r^j$, and give each nonroot clique vertex its own root coordinate divided by $\alpha-k+2$. Let $\mathbf{v}$ be the reflected vector. Substitution into the adjacency equations gives

$$(A-\alpha I)\mathbf{u}=e\mathbf{e}_{a_L},\qquad(A-\alpha I)\mathbf{v}=e\mathbf{e}_{a_0},\qquad e=(1-r^2)r^{L-1}.$$

The positive-vector bound applied to $\mathbf{u}+\mathbf{v}$ gives

$$\alpha < \lambda_1^K \le \alpha + \frac{e}{1+r^L}.$$

Reflection fixes the positive Perron vector, so $\mathbf{u}-\mathbf{v}$ is orthogonal to it. Its Rayleigh quotient, using $\|\mathbf{u}-\mathbf{v}\|_2^2 \ge 2(1-r^L)^2$, gives

$$\lambda_2^K \ge \alpha - \frac{2e(1-r^L)}{\|\mathbf{u}-\mathbf{v}\|_2^2} \ge \alpha - \frac{e}{1-r^L}.$$

Consequently

$$\delta(\mathsf{K}(k,L)) \le 2h, \qquad \lambda_1^K, \lambda_2^K \in [\alpha-2h, \alpha+2h].$$

Since $h \le r^{L-1} \le (k-2)^{-(L-1)}$, this proves (10.12).

We shall use the uniform estimate

$$L(k-2)^{-(L-1)} \le k(k-2)^{-(k-1)} \le k^{-20} \qquad (L \ge k \ge 1000). \tag{10.14}$$

The expression decreases with integral $L \ge k$, and $k-2 \ge \sqrt{k}$ proves the second inequality. It follows from (10.10) that $\lambda_2^K > k-1$ and that $z-k+2 \le 1+3/k^2$ on $[\alpha-2h, \alpha+2h]$. For $Q = F_{K_k} - r$, we therefore have

$$Q'(z) \ge F'_{K_k}(z) = 1 + \frac{k-1}{(z-k+2)^2} \ge k-1 \tag{10.15}$$

throughout this interval.

After deleting both roots, the remaining components are two $(k-1)$-cliques and the internal path. Their spectral radii are below $k-1 < \lambda_2^K$. The two-root Schur complement and reflection symmetry therefore identify the two eigenvalues by

$$\begin{aligned} Q(\lambda_1^K) &= h_L(\lambda_1^K)(1 - r(\lambda_1^K)^L), \\ Q(\lambda_2^K) &= -h_L(\lambda_2^K)(1 + r(\lambda_2^K)^L). \end{aligned}$$

Indeed the symmetric root vector belongs to the larger eigenvalue; the Schur matrix has inertia $(\text{negative}, 0)$ at the second, so its null vector there is antisymmetric. For $z \ge k-1$, differentiation of (10.3) gives

$$0 < -\frac{d}{dz}\log h_L(z) = \frac{(L-1) - 2r(z)^2/(1-r(z)^2) + 2Lr(z)^{2L}/(1-r(z)^{2L})}{\sqrt{z^2-4}} \le \frac{4L}{k}. \tag{10.16}$$

In particular, (10.14) implies

$$\frac{h_L(\lambda_2^K)}{h_L(\alpha)} \le \exp(8Lh/k) \le 1 + \frac{1}{50k}, \qquad r(\lambda_2^K)^L \le (k-2)^{-L} \le \frac{1}{100k}.$$

Integrating (10.15) between the two eigenvalues, and using that h_L decreases, yields

$$\delta(\mathsf{K}(k,L)) \le \frac{2h_L(\lambda_2^K)(1+r(\lambda_2^K)^L)}{k-1} \le \frac{2h_L(\alpha_K)}{k-2}.$$

This proves (10.11). □

Proof of Proposition 10.1. Let $G = \mathcal{D}(H_1, H_2; L)$ and suppose, towards a contradiction, that $\delta = \delta(G) \le \delta(\mathsf{K}(k, L))$. Write $\alpha_i = \alpha_{H_i}$, let M_i be the number of missing edges in H_i, and put $M = \max(M_1, M_2) \ge 1$. By (10.12) and (10.14),

$$\delta \le 2(k-2)^{-(L-1)}, \qquad L\delta \le 2k^{-20}. \tag{10.17}$$

In this proof λ_1, λ_2 are the two largest eigenvalues of G.

Since $\lambda_1 \ge \rho(H_i)$, the root-deletion estimate gives $\lambda_2 > \rho(H_i - a_i)$ and $\lambda_2 > 2$. Deleting both roots thus leaves a matrix with spectral radius below λ_2. Two-vertex interlacing implies $\lambda_3 < \lambda_2$, so the second eigenvalue is simple. On $[\lambda_2, \lambda_1]$, elimination of all nonroot vertices gives

$$\mathcal{F}(z) = \begin{pmatrix} F_{H_1}(z) - s_L(z) & -h_L(z) \\ -h_L(z) & F_{H_2}(z) - s_L(z) \end{pmatrix}. \tag{10.18}$$

Its inertia is $(\text{negative}, 0)$ at λ_2 and $(0, \text{positive})$ at λ_1. Furthermore $\mathcal{F}'(z) \succeq I$. The path contribution to this derivative has norm at most $(\lambda_2 - 2)^{-2}$: the attachment columns at the two distinct ends of the internal path are orthonormal, and the path resolvent has norm at most $(\lambda_2 - 2)^{-1}$.

If $z \ge \alpha_i$, then $F'_{H_i}(z) \le 4k$ by (10.2) and (10.6). If $z < \alpha_i$, we still have

$$z \ge \lambda_2 \ge \rho(H_i) - \delta \ge \alpha_i - 2/k - \delta.$$

This lies above $\alpha_i - 1/16$. Integrating (10.7) gives $F'_{H_i}(z) \le 4k \exp(32(2/k + \delta)) < 8k$. Including the path contribution, we obtain

$$I \preceq \mathcal{F}'(z) \preceq 10kI \qquad (\lambda_2 \le z \le \lambda_1). \tag{10.19}$$

Lemma 2.4 therefore gives

$$0 \le \mathcal{F}_{ii}(\lambda_1) \le 10k\delta, \qquad h_L(\lambda_1) \le 5k\delta.$$

Using (10.3), we find

$$-h_L(\lambda_1) r(\lambda_1)^L \le F_{H_i}(\lambda_1) - r(\lambda_1) \le 10k\delta.$$

The function $F_{H_i} - r$ has derivative at least one and zero α_i. Hence

$$|\lambda_1 - \alpha_i| \le 10k\delta \qquad (i = 1, 2). \tag{10.20}$$

Thus proximity of the two rooted parameters follows from the hypothetical small gap; it has not been assumed.

Every $z \in [\lambda_2, \lambda_1]$ is within $11k\delta$ of each α_i. Since $M_i \le k^2/20$, $k + 12M_i + 6 \le k^2$. Equations (10.6) and (10.7) show that the increase in either scalar derivative from its value at α_i is at most

$$k^2(\exp(352k\delta) - 1) < 1,$$

where the last inequality follows from (10.17). The path contribution is also less than one. It follows that

$$\mathcal{F}'(z) \preceq (k + 20(M+1))I. \tag{10.21}$$

A second application of Lemma 2.4 gives

$$\delta \ge \frac{2h_L(\lambda_1)}{k + 20(M+1)}. \tag{10.22}$$

Apply (10.5) to a core with M missing edges. Together with (10.20), this gives

$$\lambda_1 \le \alpha_K - \frac{M}{4k} + 10k\delta < \alpha_K, \tag{10.23}$$

since $10k\delta < 1/(8k) \le M/(8k)$. For $z \in [\lambda_1, \alpha_K]$,

$$-\frac{d}{dz}\log r(z) = \frac{1}{\sqrt{z^2-4}} \ge \frac{1}{k},$$

as $\alpha_K < k$. Also $r(\lambda_1) \le 2/k$. In the ratio $h_L(\lambda_1)/h_L(\alpha_K)$, the factor involving $1 - r^{2L}$ is at least one, while the logarithm of the ratio of the $1 - r^2$ factors is at least $-20/k^2$. Therefore

$$\begin{aligned}\log \frac{h_L(\lambda_1)}{h_L(\alpha_K)} &\ge \frac{(L-1)(\alpha_K - \lambda_1)}{k} - \frac{20}{k^2} \\ &\ge \frac{M(L-1)}{4k^2} - 10L\delta - \frac{20}{k^2}.\end{aligned}$$

Combining (10.11) and (10.22), we conclude that

$$\begin{aligned}\log \frac{\delta(G)}{\delta(\mathsf{K}(k,L))} &\ge \frac{M(L-1)}{4k^2} - 10L\delta - \frac{20}{k^2} - \log\frac{k + 20(M+1)}{k-2} \\ &\ge \frac{M(L-1)}{4k^2} - 10L\delta - \frac{20(M+1)+3}{k} - \frac{20}{k^2}.\end{aligned}$$

Here we used

$$\log\frac{k+20(M+1)}{k-2} = \log\left(1 + \frac{20(M+1)}{k}\right) - \log\left(1 - \frac{2}{k}\right) \le \frac{20(M+1)+3}{k}.$$

Finally, $L - 1 \ge 9999k$ and $L\delta \le 2k^{-20}$ show that the last lower bound is at least

$$\frac{9919M - 92}{4k} - \frac{20}{k^2} - 20k^{-20} > \frac{2400M}{k} > 0.$$

This contradicts $\delta(G) \le \delta(\mathsf{K}(k,L))$ and proves the proposition. □

11 Proof of the graph theorem

Proof of Theorem 1.2. Suppose there are non-double-kite minimizers at arbitrarily large orders, and choose a sequence of them. Apply the preceding graph reductions. With their notation, the graph $K = \mathsf{K}(k, L+b+q)$ has the same order as G. The sharp clique benchmark in Lemma 10.3 gives

$$\delta(K) \le \frac{2}{k}(1 + o(1))\beta_k^{-(L+b+q-1)}, \qquad \beta_k = k - 1 + O(1/k).$$

Dividing the transmission bound (9.3) by this estimate, and taking logarithms, gives

$$\begin{aligned}\log\frac{\delta(G)}{\delta(K)} \ge\ & \log\theta^2 + \log(d_1 d_2) + (b+q)\log\beta_k \\ & - (L-1)\log(\lambda_1/\beta_k) + o(1).\end{aligned} \tag{11.1}$$

The spectral bound (9.7) implies

$$\log(\lambda_1/\beta_k) \le \frac{\min(\sigma_1, q+\sigma_2)}{k} + O(1/k^2).$$

Since $L/k = 2\log k + o(\log k)$, and $b + q = O(1)$, it follows that

$$\log \frac{\delta(G)}{\delta(K)} \geq \log \theta^2 + \log(d_1 d_2) + (b+q)\log k \\ - (2\log k + o(\log k)) \min(\sigma_1, q + \sigma_2) + o(1). \tag{11.2}$$

In particular the $O(1/k)$ error in the spectral estimate contributes only $O(\log k/k) = o(1)$ here.

Pass to a subsequence on which the bounded integers b, q are fixed. Label the b residual vertices outside the shortest core path in any order. Passing to a further subsequence, all their attachment densities, the two endpoint densities d_i/k, and the two numbers $\log d_i/\log k$ converge. Write

$$d_i/k \to t_i, \qquad \frac{\log d_i}{\log k} \to \alpha_i \in [0,1], \qquad \sigma_i \to t_i^2 + H_i,$$

where H_i is the sum of squared limiting densities of off-path vertices attached to core i. If $t_i > 0$, then $\alpha_i = 1$. By (9.6), each limiting density is at most θ. Each off-path vertex meets at most one core, so $H_1 + H_2 \leq b\theta^2$.

The limiting coefficient of $\log k$ on the right of (11.2) is

$$P = b + q + \alpha_1 + \alpha_2 - 2\min(t_1^2 + H_1, q + t_2^2 + H_2).$$

Using $2\min(u, v) \leq u + v$ gives the useful decomposition

$$P \geq b(1-\theta^2) + (\alpha_1 - t_1^2) + (\alpha_2 - t_2^2). \tag{11.3}$$

Each summand on the right is nonnegative. If $t_i = 0$, its contribution is α_i; if $t_i > 0$, it is $1 - t_i^2 \geq 1 - \theta^2 > 0$. Consequently $P > 0$ if $b > 0$, if either $t_i > 0$, or if either $\alpha_i > 0$. In any of these cases (11.2) tends to $+\infty$, contradicting minimality, which requires the logarithmic ratio to be nonpositive. If all these quantities vanish but $q > 0$, then $H_1 = H_2 = 0$ and the exact expression gives $P = q > 0$, again a contradiction.

The only possible remaining subsequence therefore has $b = q = 0$ and $\alpha_1 = \alpha_2 = 0$. Now $d_i = k^{o(1)}$. There are no off-path vertices, so $\sigma_i = d_i^2/k^2 = k^{-2+o(1)}$; its entire logarithmic penalty in (11.2) is $o(1)$. We conclude that

$$\frac{\delta(G)}{\delta(K)} \geq \theta^2 d_1 d_2 (1 - o(1)).$$

If the integer product $d_1 d_2$ is at least two on a further subsequence, this exceeds one because $2\theta^2 > 1$. Thus $d_1 = d_2 = 1$ eventually.

Every residual vertex now lies on the shortest path between the two cores. That path is induced; shortestness excludes every other core-to-path edge. The two endpoint attachments are single edges, the cores have equal order k, and there are no edges between the cores. Thus $G = \mathcal{D}(G[C_1], G[C_2]; L)$ in the notation of Section 10. Lemma 8.1 gives the required core minimum degrees. Since $L/(k\log k) \to 2$, eventually $k \geq 1000$ and $L \geq 10000k$. Proposition 10.1 forces both cores to be complete, contradicting the choice of the sequence.

The same reasoning applies to any sequence of exceptional minimizers. Their existence at unbounded orders is therefore impossible, proving the existence of one $N_{\mathcal{G}}$ valid for every minimizer. The parameter asymptotics follow from $k = k_0 + O(1)$, (7.4), and $n = 2k + L - 1$. □

12 Scope of the conclusions

Theorems 1.1 and 1.2 identify every minimizing graph in their respective classes at all sufficiently large orders. The proofs give existence of thresholds and do not determine the remaining finite ranges. They also leave open whether the optimizing integer parameter is unique at every sufficiently large order. All comparisons above are between graphs of the same order; none of the structural conclusions requires a numerical enumeration.

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